The Switchback Law

The prime-angle sign sequence is scale-invariant — the same seesaw in the hundreds and the millions

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What this is. An empirical regularity with a clean lens and one open thread. The Prime-Triangle angle wobbles under its 45° ceiling; strip away the *size* of the wobble and keep only the **sign** of each change (up or down), and that sign-sequence — how often it flips, how long same-direction runs last — is **the same at every scale**, even as the wobble itself shrinks by orders of magnitude. It also yields a usable **betting rule** for the next sign, and its fine structure is the wheel. Tags: **[emp]** measured · **[null]** shuffle-controlled · **[interp]** placement · **[resolved]** the mod6 dip = leading-order Hardy–Littlewood, matched to 0.1 point (§5). Companion: *The Prime-Triangle Angle* (the wobble and its ceiling); *Prime-Gap Memory and the Differencing Trap* (the two-null discipline used here).

Placement up front (no inflation): this is a fresh, visual face of something number theory already expects — the ***scale-invariance of the normalized local statistics of primes*** *(the Cramér / Hardy–Littlewood picture). It is not a new theorem. Its value is the lens (it makes that invariance graspable and turns it into a rule you can use) and the* mod6 *fine structure (§5). It sits alongside the other shadows in this collection, not above them.*

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1. The object: switchbacks

For consecutive primes, the Prime-Triangle angle is $\alpha_n = \arctan(p_n/p_{n+1}) \approx 45° - \frac{90}{\pi}\, g_n/p_{n+1}$ — a rescaled gap, pinned just under 45°. Its **change** $\Delta\alpha_n = \alpha_n - \alpha_{n-1}$ carries a sign: **up** (the gap shrank, the angle rose toward the ceiling) or **down** (the gap grew). Reading only that sign, the sequence is a string of switchbacks: $- - + - + + - \ldots$

On the ordinary number line this sign-structure is invisible — "this gap, then subtract that gap, then the next" has no picture. The triangle is what renders it; that is why the pattern is *seeable* at all.

2. The run-length law

Same-direction runs are short, and they die **faster than a coin.** Over the primes in $[10^6, 3 \times 10^6]$:

run of length	share of runs	a fair coin would give
1	63.1%	50%
2	27.3%	25%
3	7.7%	12.5%
4	1.6%	6.25%
5	0.3%	3.1%
6–7	0.07%	1.6%

[emp] Each extra same-sign step is ~4–5× less likely than the last — the sequence *over-alternates* relative to random. Equivalently, the **reversal probability climbs with run length**:

after a run of…	the next change flips
1	63%
2	74%
3	80%
4	82%

after a run of...	the next change flips

So the sign is genuinely **predictable**: once you have seen two or three moves the same way, bet on a reversal and you are right three or four times out of five. (This is the discipline the wobble note's "runs $\approx 1/3$" constant pointed at, now read as a usable table.)

3. The law is scale-invariant — the headline

The *amplitude* of the wobble collapses with scale — $\mathrm{RMS}(\Delta\alpha) \approx 33 \ln p/p$, four orders of magnitude smaller near 10^6 than near 10^2. The **sign-structure does not move at all:**

run length	1	2	3	4	5	6
hundreds (first 1000 primes)	62.2%	29.5%	6.9%	1.3%	—	—
millions ($[10^6, 3 \times 10^6]$)	63.1%	27.3%	7.7%	1.6%	0.3%	0.06%

The two distributions are the same. In the first thousand primes there are **exactly nine** runs of length four and none longer; in the millions the identical *rate* produces occasional 5s, 6s, a rare 7 — so "four is the maximum" is a sample-size floor, not a hard ceiling, but the **law generating it is fixed across scale.** Zoom into the millions and the switchbacks look exactly like the hundreds. *This is the point of the note.*

4. Why the runs die fast: the ceiling, felt from the changes

The over-alternation is the boundedness of the angle seen from the change side. Because primes increase with bounded gaps, α is trapped in a shrinking band below 45°; it cannot drift, so its changes cannot run the same way for long without being pulled back — an **overcorrection** that always returns the angle to the band (and, being a return, tends to overshoot, seeding the next reversal). That this is structural and not generic noise is shown in the companion note by the two-null test: shuffle the *gaps* and the band-and-cancellation survive; shuffle the *changes* and let them free-float, and the reconstructed angle wanders straight through 45°. The switchback tightness is held in place by the primes being a real monotone, bounded-gap sequence.

5. The fine structure is the wheel: a $\bmod 6$ suppression

Within the sign-law lives an arithmetic fingerprint. Look at the size of the gap-change, $|\Delta g| = |g_{n+1} - g_n|$, over $[10^6, 3 \times 10^6]$:

$\lvert\Delta g\rvert$	2	4	**6**	8	10	**12**	14
frequency	16.8%	13.3%	**5.1%**	11.4%	8.8%	**3.0%**	6.6%

[emp] The multiples of 6 are **suppressed** — sharp local minima at 6 and 12. Since gaps are even, $|\Delta g| \equiv 0 \pmod 6$ means two consecutive gaps are *congruent* mod 6; the dip says **consecutive gaps avoid repeating their residue** $\bmod 6$**.** Quantitatively the avoidance is nearly 2-to-1: the same-residue rate is 0.176 against 0.348 if consecutive gaps were independent.

Does Hardy–Littlewood predict it? Yes — to the digit. Build the leading-order singular-series local model (the wheel-Cramér surrogate: small-prime divisibility exact, everything else independent random) and read off its $|\Delta g|$ distribution:

$\lvert\Delta g\rvert$	4	**6**	8	10	**12**	14
real primes $> 10^6$	12.70%	**5.09%**	10.94%	8.62%	**2.99%**	6.62%
wheel model	12.66%	**5.09%**	10.85%	8.59%	**3.01%**	6.63%

The model reproduces the whole distribution — **including the dip depths at 6** (5.09 **vs** 5.09) **and 12** (2.99 **vs** 3.01) — to within 0.1 point. So the mod6 suppression is a **leading-order Hardy–Littlewood effect**, exactly: the singular series forces consecutive gaps to avoid a shared residue, and the wheel model that encodes it lands on the measured depths.

[**resolved, with a small tail**] On top of the leading term sits a **Lemke Oliver–Soundararajan lower-order correction.** The same-residue bias runs slightly deeper at small scale and fades toward the wheel's asymptotic value — -0.193 at 10^4, -0.180 at 10^5, -0.174 at 10^6, -0.171 at 10^7 — the characteristic slow ($\sim 1/\log$) decay of the LOS secondary bias (the "unexpected biases in consecutive primes," 2016). So the dip is Hardy–Littlewood to leading order (matched to the digit) plus a decaying LOS tail. The thread closes where it should: it *is* the wheel.

6. Where it lands, and what it's good for

Prediction. The betting table of §2 is scale-invariant (§3), so it *transfers*: hunting the sign-rhythm around a record prime, the wobble sizes up there are meaningless, but the run-length odds you learned in the hundreds are exactly the odds in the billions. Combined with the Seven-Sisters generator ($2p + k$ for the wheel-strong offsets), this is the honest content of "how to guess where the next prime leans" — not a formula, a *rhythm with known odds.*

Placement, plainly. The switchback law is a real, clean, scale-invariant regularity — but it is a *consequence* of the conjectured scale-invariance of normalized prime local statistics, not a new fact about the deep structure of primes. Its contribution is the **lens** (it makes that invariance visible and usable, through the angle sign-sequence that the number line hides) and the mod6 **fine structure** (§5), which ties it back to the wheel. It is one more shadow — the seesaw one — not a summit.

Appendix. Reproduction

```python
import math, bisect
from collections import Counter, defaultdict
def sieve(n):
    b=bytearray([1])*(n+1); b[0]=b[1]=0; i=2
    while i*i<=n:
        if b[i]: b[i*i::i]=bytearray(len(b[i*i::i]))
        i+=1
    return b
B=sieve(3_000_000); allp=[i for i in range(2,3_000_000) if B[i]]
def runs_and_reversal(P):
    a=[math.degrees(math.atan(P[i]/P[i+1])) for i in range(len(P)-1)]
    da=[a[i+1]-a[i] for i in range(len(a)-1)]
    s=[1 if d>0 else -1 if d<0 else 0 for d in da]
    runlen=Counter(); i=0
    while i<len(s):
        j=i
        while j<len(s) and s[j]==s[i] and s[i]!=0: j+=1
        if s[i]!=0: runlen[j-i]+=1
        i=max(j,i+1)
```

```
    return runlen
i1=bisect.bisect_left(allp,1_000_000); i2=bisect.bisect_left(allp,3_000_000)
print('hundreds:', runs_and_reversal(allp[:1000]))
print('millions:', runs_and_reversal(allp[i1:i2]))
# mod-6 suppression:
g=[allp[i+1]-allp[i] for i in range(i1,i2)]
print(Counter(abs(g[i+1]-g[i]) for i in range(len(g)-1)).most_common(8))
```

References

- A. Proxmire, *The Prime-Triangle Angle* (this collection) — the wobble, the 45° ceiling, the $33 \ln p/p$ envelope.
- A. Proxmire, *Prime-Gap Memory and the Differencing Trap* (this collection) — the two-null discipline (shuffle gaps vs shuffle changes) used in §4.
- R. J. Lemke Oliver, K. Soundararajan, *Unexpected biases in the distribution of consecutive primes* (PNAS, 2016) — the residue-correlation mechanism behind the mod 6 suppression of §5.
- Empirical basis: *Prime Trigonometry Data* (author's spreadsheet, first 1000 pairs) cross-checked against a sieve to 3×10^6.